

TERRAFORM

Assignment : 1

Tasks To Be Performed:

1. Create an EC2 service in the default subnet in the Ohio region

```
PS C:\Users\jayan\Terraform\Terraform_practice> terraform init
```

Initializing the backend...

Initializing provider plugins...

- Reusing previous version of hashicorp/aws from the dependency lock file

- Using previously-installed hashicorp/aws v6.50.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
PS C:\Users\jayan\Terraform\Terraform_practice> terraform validate
```

Success! The configuration is valid.

```
PS C:\Users\jayan\Terraform\Terraform_practice> terraform plan
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the

following symbols:

- + create

Terraform will perform the following actions:

aws_instance.myec2 will be created

```
+ resource "aws_instance" "myec2" {  
  
  + ami                = "ami-0741dc526e1106ae5"  
  
  + arn                = (known after apply)  
  
  + associate_public_ip_address    = (known after apply)  
  
  + availability_zone    = (known after apply)  
  
  + disable_api_stop    = (known after apply)  
  
  + disable_api_termination    = (known after apply)  
  
  + ebs_optimized        = (known after apply)  
  
  + enable_primary_ipv6    = (known after apply)  
  
  + force_destroy        = false  
  
  + get_password_data    = false  
  
  + host_id              = (known after apply)  
  
  + host_resource_group_arn    = (known after apply)  
  
  + iam_instance_profile    = (known after apply)  
  
  + id                  = (known after apply)  
  
  + instance_initiated_shutdown_behavior = (known after apply)  
  
  + instance_lifecycle    = (known after apply)  
  
  + instance_state        = (known after apply)  
  
  + instance_type        = "t3.micro"  
  
  + ipv6_address_count    = (known after apply)  
  
  + ipv6_addresses        = (known after apply)  
  
  + key_name              = (known after apply)  
  
  + monitoring            = (known after apply)  
}
```

+ outpost_arn = (known after apply)

+ password_data = (known after apply)

+ placement_group = (known after apply)

+ placement_group_id = (known after apply)

+ placement_partition_number = (known after apply)

+ primary_network_interface_id = (known after apply)

+ private_dns = (known after apply)

+ private_ip = (known after apply)

+ public_dns = (known after apply)

+ public_ip = (known after apply)

+ region = "us-east-2"

+ secondary_private_ips = (known after apply)

+ security_groups = (known after apply)

+ source_dest_check = true

+ spot_instance_request_id = (known after apply)

+ subnet_id = (known after apply)

+ tags = {

 + "Name" = "Terraform-EC2"

}

+ tags_all = {

 + "Name" = "Terraform-EC2"

}

+ tenancy = (known after apply)

+ user_data_base64 = (known after apply)

+ user_data_replace_on_change = false

+ vpc_security_group_ids = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ primary_network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)

+ secondary_network_interface (known after apply)

}

Plan: 1 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you

run "terraform apply" now.

PS C:\Users\jayan\Terraform\Terraform_practice> terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the

following symbols:

+ create

Terraform will perform the following actions:

aws_instance.myec2 will be created

+ resource "aws_instance" "myec2" {

+ ami = "ami-0741dc526e1106ae5"

+ arn = (known after apply)

+ associate_public_ip_address = (known after apply)

+ availability_zone = (known after apply)

+ disable_api_stop = (known after apply)

+ disable_api_termination = (known after apply)

+ ebs_optimized = (known after apply)

+ enable_primary_ipv6 = (known after apply)

+ force_destroy = false

+ get_password_data = false

+ host_id = (known after apply)

+ host_resource_group_arn = (known after apply)

+ iam_instance_profile = (known after apply)

+ id = (known after apply)

+ instance_initiated_shutdown_behavior = (known after apply)

+ instance_lifecycle = (known after apply)

+ instance_state = (known after apply)

+ instance_type = "t3.micro"

+ ipv6_address_count = (known after apply)

+ ipv6_addresses = (known after apply)

+ key_name = (known after apply)

+ monitoring = (known after apply)

+ outpost_arn = (known after apply)

+ password_data = (known after apply)

+ placement_group = (known after apply)

+ placement_group_id = (known after apply)

+ placement_partition_number = (known after apply)

+ primary_network_interface_id = (known after apply)

+ private_dns = (known after apply)

+ private_ip = (known after apply)

+ public_dns = (known after apply)

+ public_ip = (known after apply)

+ region = "us-east-2"

+ secondary_private_ips = (known after apply)

+ security_groups = (known after apply)

+ source_dest_check = true

+ spot_instance_request_id = (known after apply)

+ subnet_id = (known after apply)

+ tags = {
 + "Name" = "Terraform-EC2"
}

+ tags_all = {
 + "Name" = "Terraform-EC2"
}

+ tenancy = (known after apply)

+ user_data_base64 = (known after apply)

+ user_data_replace_on_change = false

+ vpc_security_group_ids = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ primary_network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)

+ secondary_network_interface (known after apply)

}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

aws_instance.myec2: Creating...

aws_instance.myec2: Still creating... [00m10s elapsed]

aws_instance.myec2: Creation complete after 18s [id=i-0ce1641f15504027b]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

PS C:\Users\jayan\Terraform\Terraform_practice> terraform show


```
# aws_instance.myec2:
```

```
resource "aws_instance" "myec2" {

  ami                = "ami-0741dc526e1106ae5"

  arn                = "arn:aws:ec2:us-east-2:931628309022:instance/i-0ce1641f15504027b"

  associate_public_ip_address    = true

  availability_zone              = "us-east-2c"

  disable_api_stop              = false

  disable_api_termination       = false

  ebs_optimized                 = false

  force_destroy                 = false

  get_password_data             = false

  hibernation                   = false

  host_id                      = null

  iam_instance_profile          = null

  id                           = "i-0ce1641f15504027b"

  instance_initiated_shutdown_behavior = "stop"

  instance_lifecycle           = null

  instance_state               = "running"

  instance_type                = "t3.micro"

  ipv6_address_count            = 0

  ipv6_addresses               = []

  key_name                     = null

  monitoring                   = false

  outpost_arn                  = null

  password_data                = null

  placement_group              = null

  placement_group_id           = null
```

```
placement_partition_number      = 0

primary_network_interface_id    = "eni-08a5553559c314fbe"

private_dns                     = "ip-172-31-47-252.us-east-2.compute.internal"

private_ip                      = "172.31.47.252"

public_dns                      = "ec2-18-221-254-150.us-east-2.compute.amazonaws.com"

public_ip                       = "18.221.254.150"

region                          = "us-east-2"

secondary_private_ips           = []

security_groups                 = [

    "default",

]

source_dest_check               = true

spot_instance_request_id       = null

subnet_id                      = "subnet-035dd2f302382ecf7"

tags                            = {

    "Name" = "Terraform-EC2"

}

tags_all                        = {

    "Name" = "Terraform-EC2"

}

tenancy                         = "default"

user_data_replace_on_change     = false

vpc_security_group_ids         = [

    "sg-0c68c12a93a123604",

]

capacity_reservation_specification {
```

```
capacity_reservation_preference = "open"

}
```

```
cpu_options {

  amd_sev_snp      = null

  core_count      = 1

  nested_virtualization = null

  threads_per_core  = 2

}
```

```
credit_specification {

  cpu_credits = "unlimited"

}
```

```
enclave_options {

  enabled = false

}
```

```
maintenance_options {

  auto_recovery = "default"

}
```

```
metadata_options {

  http_endpoint      = "enabled"

  http_protocol_ipv6  = "disabled"

  http_put_response_hop_limit = 2

  http_tokens        = "required"

}
```

```
instance_metadata_tags    = "disabled"

}

primary_network_interface {

    delete_on_termination = true

    network_interface_id = "eni-08a5553559c314fbe"

}

private_dns_name_options {

    enable_resource_name_dns_a_record  = false

    enable_resource_name_dns_aaaa_record = false

    hostname_type                      = "ip-name"

}

root_block_device {

    delete_on_termination = true

    device_name           = "/dev/xvda"

    encrypted             = false

    iops                  = 3000

    kms_key_id            = null

    tags                  = {}

    tags_all              = {}

    throughput            = 125

    volume_id             = "vol-08f0b356969dc5067"

    volume_size           = 8

    volume_type           = "gp3"

}
```

```

}

PS C:\Users\jayan\Terraform\Terraform_practice>

=====

ec.tf:

provider "aws" {

    region = "us-east-2"

}

resource "aws_instance" "myec2" {

    ami      = "ami-0741dc526e1106ae5" # Amazon Linux 2 AMI

    instance_type = "t3.micro"

    tags = {

        Name = "Terraform-EC2"

    }

}

```

Assignment : 2

Tasks To Be Performed:

1. Destroy the previous deployment

PS C:\Users\jayan\Terraform\Terraform_practice> terraform destroy

aws_instance.myec2: Refreshing state... [id=i-0ce1641f15504027b]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the

following symbols:

- destroy

Terraform will perform the following actions:

aws_instance.myec2 will be destroyed

```
- resource "aws_instance" "myec2" {  
  
  - ami                  = "ami-0741dc526e1106ae5" -> null  
  
  - arn                  = "arn:aws:ec2:us-east-2:931628309022:instance/i-0ce1641f15504027b" -> null  
  
  - associate_public_ip_address    = true -> null  
  
  - availability_zone              = "us-east-2c" -> null  
  
  - disable_api_stop               = false -> null  
  
  - disable_api_termination        = false -> null  
  
  - ebs_optimized                  = false -> null  
  
  - force_destroy                  = false -> null  
  
  - get_password_data              = false -> null  
  
  - hibernation                    = false -> null  
  
  - id                            = "i-0ce1641f15504027b" -> null  
  
  - instance_initiated_shutdown_behavior = "stop" -> null  
  
  - instance_state                 = "running" -> null  
  
  - instance_type                  = "t3.micro" -> null  
  
  - ipv6_address_count              = 0 -> null  
  
  - ipv6_addresses                  = [] -> null  
  
  - monitoring                     = false -> null  
  
  - placement_partition_number      = 0 -> null  
  
  - primary_network_interface_id    = "eni-08a5553559c314fbe" -> null  
  
  - private_dns                     = "ip-172-31-47-252.us-east-2.compute.internal" -> null  
  
  - private_ip                      = "172.31.47.252" -> null  
  
  - public_dns                      = "ec2-18-221-254-150.us-east-2.compute.amazonaws.com" -> null  
  
  - public_ip                       = "18.221.254.150" -> null
```

```
- region                = "us-east-2" -> null

- secondary_private_ips    = [] -> null

- security_groups          = [

  - "default",

] -> null

- source_dest_check        = true -> null

- subnet_id                = "subnet-035dd2f302382ecf7" -> null

- tags                     = {

  - "Name" = "Terraform-EC2"

} -> null

- tags_all                 = {

  - "Name" = "Terraform-EC2"

} -> null

- tenancy                  = "default" -> null

- user_data_replace_on_change = false -> null

- vpc_security_group_ids    = [

  - "sg-0c68c12a93a123604",

] -> null

# (9 unchanged attributes hidden)

- capacity_reservation_specification {

  - capacity_reservation_preference = "open" -> null

}

- cpu_options {

  - core_count      = 1 -> null

  - threads_per_core = 2 -> null
```

(2 unchanged attributes hidden)

}

- credit_specification {

- cpu_credits = "unlimited" -> null

}

- enclave_options {

- enabled = false -> null

}

- maintenance_options {

- auto_recovery = "default" -> null

}

- metadata_options {

- http_endpoint = "enabled" -> null

- http_protocol_ipv6 = "disabled" -> null

- http_put_response_hop_limit = 2 -> null

- http_tokens = "required" -> null

- instance_metadata_tags = "disabled" -> null

}

- primary_network_interface {

- delete_on_termination = true -> null

- network_interface_id = "eni-08a5553559c314fbe" -> null

}


```
- private_dns_name_options {  
  
  - enable_resource_name_dns_a_record  = false -> null  
  
  - enable_resource_name_dns_aaaa_record = false -> null  
  
  - hostname_type                      = "ip-name" -> null  
  
}  
  
- root_block_device {  
  
  - delete_on_termination = true -> null  
  
  - device_name          = "/dev/xvda" -> null  
  
  - encrypted            = false -> null  
  
  - iops                  = 3000 -> null  
  
  - tags                  = {} -> null  
  
  - tags_all              = {} -> null  
  
  - throughput            = 125 -> null  
  
  - volume_id             = "vol-08f0b356969dc5067" -> null  
  
  - volume_size           = 8 -> null  
  
  - volume_type           = "gp3" -> null  
  
  # (1 unchanged attribute hidden)  
  
}  
  
}
```

Plan: 0 to add, 0 to change, 1 to destroy.

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.

There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

aws_instance.myec2: Destroying... [id=i-0ce1641f15504027b]

aws_instance.myec2: Still destroying... [id=i-0ce1641f15504027b, 00m10s elapsed]

aws_instance.myec2: Still destroying... [id=i-0ce1641f15504027b, 00m20s elapsed]

aws_instance.myec2: Destruction complete after 22s

Destroy complete! Resources: 1 destroyed.

PS C:\Users\jayan\Terraform\Terraform_practice>

2. Create a new EC2 instance with an Elastic IP

PS C:\Users\jayan\Terraform\Terraform_practice> terraform validate

Success! The configuration is valid.

PS C:\Users\jayan\Terraform\Terraform_practice> terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the

following symbols:

- + create

Terraform will perform the following actions:

aws_eip.myec2 will be created

+ resource "aws_eip" "myec2" {

- + allocation_id = (known after apply)

- + arn = (known after apply)

- + association_id = (known after apply)

```
+ carrier_ip      = (known after apply)

+ customer_owned_ip = (known after apply)

+ domain          = "vpc"

+ id              = (known after apply)

+ instance        = (known after apply)

+ ipam_pool_id     = (known after apply)

+ network_border_group = (known after apply)

+ network_interface = (known after apply)

+ private_dns      = (known after apply)

+ private_ip       = (known after apply)

+ ptr_record       = (known after apply)

+ public_dns       = (known after apply)

+ public_ip        = (known after apply)

+ public_ipv4_pool = (known after apply)

+ region          = "us-east-2"

+ tags_all        = (known after apply)

}
```

aws_instance.myec2 will be created

```
+ resource "aws_instance" "myec2" {

  + ami              = "ami-0741dc526e1106ae5"

  + arn              = (known after apply)

  + associate_public_ip_address = (known after apply)

  + availability_zone = (known after apply)

  + disable_api_stop  = (known after apply)

  + disable_api_termination = (known after apply)

  + ebs_optimized     = (known after apply)
```

+ enable_primary_ipv6 = (known after apply)

+ force_destroy = false

+ get_password_data = false

+ host_id = (known after apply)

+ host_resource_group_arn = (known after apply)

+ iam_instance_profile = (known after apply)

+ id = (known after apply)

+ instance_initiated_shutdown_behavior = (known after apply)

+ instance_lifecycle = (known after apply)

+ instance_state = (known after apply)

+ instance_type = "t3.micro"

+ ipv6_address_count = (known after apply)

+ ipv6_addresses = (known after apply)

+ key_name = (known after apply)

+ monitoring = (known after apply)

+ outpost_arn = (known after apply)

+ password_data = (known after apply)

+ placement_group = (known after apply)

+ placement_group_id = (known after apply)

+ placement_partition_number = (known after apply)

+ primary_network_interface_id = (known after apply)

+ private_dns = (known after apply)

+ private_ip = (known after apply)

+ public_dns = (known after apply)

+ public_ip = (known after apply)

+ region = "us-east-2"

+ secondary_private_ips = (known after apply)

+ security_groups = (known after apply)

+ source_dest_check = true

+ spot_instance_request_id = (known after apply)

+ subnet_id = (known after apply)

+ tags = {

 + "Name" = "Terraform-EC2"

}

+ tags_all = {

 + "Name" = "Terraform-EC2"

}

+ tenancy = (known after apply)

+ user_data_base64 = (known after apply)

+ user_data_replace_on_change = false

+ vpc_security_group_ids = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

```
+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ primary_network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)

+ secondary_network_interface (known after apply)
}
```

Plan: 2 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you

run "terraform apply" now.

PS C:\Users\jayan\Terraform\Terraform_practice> terraform apply -auto-approve

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the

following symbols:

+ create

Terraform will perform the following actions:

aws_eip.myeip will be created

```
+ resource "aws_eip" "myeip" {  
  
  + allocation_id    = (known after apply)  
  
  + arn              = (known after apply)  
  
  + association_id   = (known after apply)  
  
  + carrier_ip       = (known after apply)  
  
  + customer_owned_ip = (known after apply)  
  
  + domain           = "vpc"  
  
  + id               = (known after apply)  
  
  + instance         = (known after apply)  
  
  + ipam_pool_id     = (known after apply)  
  
  + network_border_group = (known after apply)  
  
  + network_interface = (known after apply)  
  
  + private_dns      = (known after apply)  
  
  + private_ip       = (known after apply)  
  
  + ptr_record       = (known after apply)  
  
  + public_dns       = (known after apply)  
  
  + public_ip        = (known after apply)  
  
  + public_ipv4_pool  = (known after apply)  
  
  + region           = "us-east-2"  
  
  + tags_all         = (known after apply)  
  
}
```

aws_instance.myec2 will be created

```
+ resource "aws_instance" "myec2" {  
  
  + ami                  = "ami-0741dc526e1106ae5"  
  
  + arn                  = (known after apply)  
  
  + associate_public_ip_address = (known after apply)  
  
  + availability_zone      = (known after apply)  
  
  + disable_api_stop      = (known after apply)  
  
  + disable_api_termination = (known after apply)  
  
  + ebs_optimized         = (known after apply)  
  
  + enable_primary_ipv6    = (known after apply)  
  
  + force_destroy         = false  
  
  + get_password_data      = false  
  
  + host_id               = (known after apply)  
  
  + host_resource_group_arn = (known after apply)  
  
  + iam_instance_profile   = (known after apply)  
  
  + id                    = (known after apply)  
  
  + instance_initiated_shutdown_behavior = (known after apply)  
  
  + instance_lifecycle     = (known after apply)  
  
  + instance_state         = (known after apply)  
  
  + instance_type          = "t3.micro"  
  
  + ipv6_address_count     = (known after apply)  
  
  + ipv6_addresses         = (known after apply)  
  
  + key_name               = (known after apply)  
  
  + monitoring             = (known after apply)  
  
  + outpost_arn            = (known after apply)  
  
  + password_data          = (known after apply)  
  
  + placement_group        = (known after apply)
```


+ placement_group_id = (known after apply)

+ placement_partition_number = (known after apply)

+ primary_network_interface_id = (known after apply)

+ private_dns = (known after apply)

+ private_ip = (known after apply)

+ public_dns = (known after apply)

+ public_ip = (known after apply)

+ region = "us-east-2"

+ secondary_private_ips = (known after apply)

+ security_groups = (known after apply)

+ source_dest_check = true

+ spot_instance_request_id = (known after apply)

+ subnet_id = (known after apply)

+ tags = {

 + "Name" = "Terraform-EC2"

}

+ tags_all = {

 + "Name" = "Terraform-EC2"

}

+ tenancy = (known after apply)

+ user_data_base64 = (known after apply)

+ user_data_replace_on_change = false

+ vpc_security_group_ids = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

```
+ ebs_block_device (known after apply)

+ enclave_options (known after apply)

+ ephemeral_block_device (known after apply)

+ instance_market_options (known after apply)

+ maintenance_options (known after apply)

+ metadata_options (known after apply)

+ network_interface (known after apply)

+ primary_network_interface (known after apply)

+ private_dns_name_options (known after apply)

+ root_block_device (known after apply)

+ secondary_network_interface (known after apply)
}
```

Plan: 2 to add, 0 to change, 0 to destroy.

aws_instance.myec2: Creating...

aws_instance.myec2: Still creating... [00m10s elapsed]

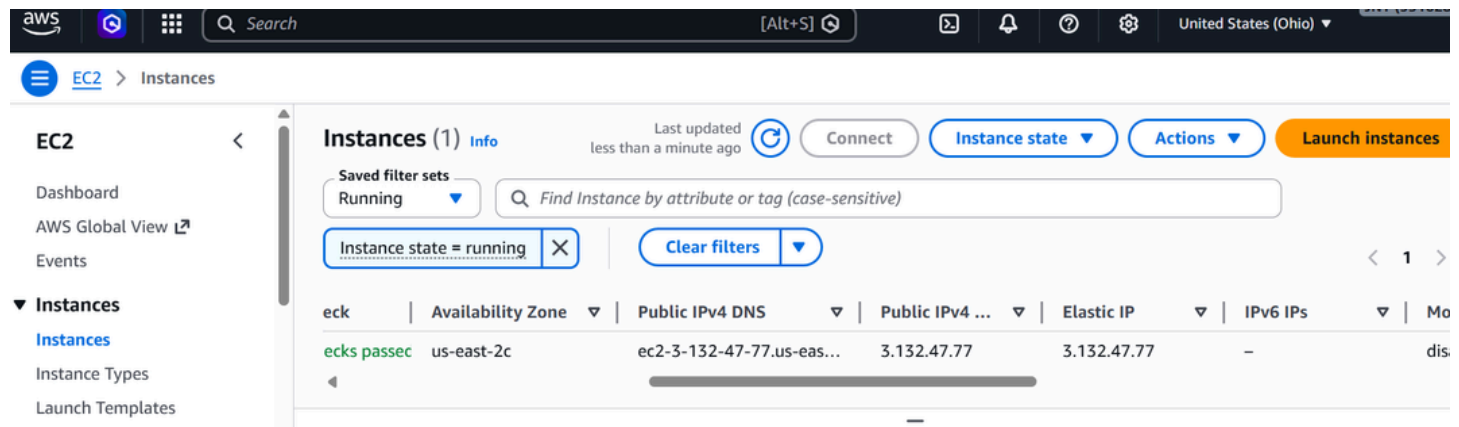
aws_instance.myec2: Creation complete after 18s [id=i-014c7b457c83c91ca]

aws_eip.myec2: Creating...

aws_eip.myec2: Creation complete after 4s [id=eipalloc-06d329a76e85c5c58]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

PS C:\Users\jayan\Terraform\Terraform_practice>



Modified ec2.tf:

```
provider "aws" {
```

```
    region = "us-east-2"
```

```
}
```

```
resource "aws_instance" "myec2" {
```

```
    ami           = "ami-0741dc526e1106ae5" # Amazon Linux 2 AMI
```

```
    instance_type = "t3.micro"
```

```
    tags = {
```

```
        Name = "Terraform-EC2"
```

```
    }
```

```
}
```

```
resource "aws_eip" "myeip" {
```

```
    instance = aws_instance.myec2.id
```

```
    domain = "vpc"
```

```
}
```

Assignment : 3

Tasks To Be Performed:

1. Destroy the previous deployment

```
PS C:\Users\jayan\Terraform\Terraform_practice> terraform destroy
```

```
aws_instance.myec2: Refreshing state... [id=i-014c7b457c83c91ca]
```

```
aws_eip.myecip: Refreshing state... [id=eipalloc-06d329a76e85c5c58]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the

following symbols:

- destroy

Terraform will perform the following actions:

```
# aws_eip.myecip will be destroyed
```

```
- resource "aws_eip" "myecip" {  
  - allocation_id      = "eipalloc-06d329a76e85c5c58" -> null  
  - arn                = "arn:aws:ec2:us-east-2:931628309022:elastic-ip/eipalloc-06d329a76e85c5c58" -> null  
  - association_id     = "eipassoc-0077192fb3b18c33f" -> null  
  - domain            = "vpc" -> null  
  - id                = "eipalloc-06d329a76e85c5c58" -> null  
  - instance          = "i-014c7b457c83c91ca" -> null  
  - network_border_group = "us-east-2" -> null  
  - network_interface  = "eni-09808a08492cfd273" -> null  
  - private_dns        = "ip-172-31-39-97.us-east-2.compute.internal" -> null
```

```

- private_ip      = "172.31.39.97" -> null

- public_dns      = "ec2-3-132-47-77.us-east-2.compute.amazonaws.com" -> null

- public_ip       = "3.132.47.77" -> null

- public_ipv4_pool = "amazon" -> null

- region          = "us-east-2" -> null

- tags            = {} -> null

- tags_all        = {} -> null

# (5 unchanged attributes hidden)

}

# aws_instance.myec2 will be destroyed

- resource "aws_instance" "myec2" {

  - ami            = "ami-0741dc526e1106ae5" -> null

  - arn            = "arn:aws:ec2:us-east-2:931628309022:instance/i-014c7b457c83c91ca" -> null

  - associate_public_ip_address = true -> null

  - availability_zone          = "us-east-2c" -> null

  - disable_api_stop          = false -> null

  - disable_api_termination    = false -> null

  - ebs_optimized              = false -> null

  - force_destroy              = false -> null

  - get_password_data          = false -> null

  - hibernation                 = false -> null

  - id                         = "i-014c7b457c83c91ca" -> null

  - instance_initiated_shutdown_behavior = "stop" -> null

  - instance_state             = "running" -> null

  - instance_type              = "t3.micro" -> null

  - ipv6_address_count         = 0 -> null

```

```
- ipv6_addresses          = [] -> null

- monitoring              = false -> null

- placement_partition_number = 0 -> null

- primary_network_interface_id = "eni-09808a08492cfd273" -> null

- private_dns             = "ip-172-31-39-97.us-east-2.compute.internal" -> null

- private_ip              = "172.31.39.97" -> null

- public_dns              = "ec2-3-132-47-77.us-east-2.compute.amazonaws.com" -> null

- public_ip               = "3.132.47.77" -> null

- region                  = "us-east-2" -> null

- secondary_private_ips    = [] -> null

- security_groups          = [

  - "default",

] -> null

- source_dest_check        = true -> null

- subnet_id               = "subnet-035dd2f302382ecf7" -> null

- tags                    = {

  - "Name" = "Terraform-EC2"

} -> null

- tags_all                 = {

  - "Name" = "Terraform-EC2"

} -> null

- tenancy                  = "default" -> null

- user_data_replace_on_change = false -> null

- vpc_security_group_ids    = [

  - "sg-0c68c12a93a123604",

] -> null

# (9 unchanged attributes hidden)
```

```
- capacity_reservation_specification {  
    - capacity_reservation_preference = "open" -> null  
}
```

```
- cpu_options {  
    - core_count      = 1 -> null  
    - threads_per_core = 2 -> null  
    # (2 unchanged attributes hidden)  
}
```

```
- credit_specification {  
    - cpu_credits = "unlimited" -> null  
}
```

```
- enclave_options {  
    - enabled = false -> null  
}
```

```
- maintenance_options {  
    - auto_recovery = "default" -> null  
}
```

```
- metadata_options {  
    - http_endpoint      = "enabled" -> null  
    - http_protocol_ipv6 = "disabled" -> null  
    - http_put_response_hop_limit = 2 -> null
```

```
- http_tokens          = "required" -> null

- instance_metadata_tags = "disabled" -> null

}

- primary_network_interface {

  - delete_on_termination = true -> null

  - network_interface_id = "eni-09808a08492cfd273" -> null

}

- private_dns_name_options {

  - enable_resource_name_dns_a_record = false -> null

  - enable_resource_name_dns_aaaa_record = false -> null

  - hostname_type = "ip-name" -> null

}

- root_block_device {

  - delete_on_termination = true -> null

  - device_name = "/dev/xvda" -> null

  - encrypted = false -> null

  - iops = 3000 -> null

  - tags = {} -> null

  - tags_all = {} -> null

  - throughput = 125 -> null

  - volume_id = "vol-090579bed5d51b921" -> null

  - volume_size = 8 -> null

  - volume_type = "gp3" -> null

  # (1 unchanged attribute hidden)
```



```
}
```

```
}
```

Plan: 0 to add, 0 to change, 2 to destroy.

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.

There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

aws_eip.myeip: Destroying... [id=eipalloc-06d329a76e85c5c58]

aws_eip.myeip: Destruction complete after 3s

aws_instance.myec2: Destroying... [id=i-014c7b457c83c91ca]

aws_instance.myec2: Still destroying... [id=i-014c7b457c83c91ca, 00m10s elapsed]

aws_instance.myec2: Still destroying... [id=i-014c7b457c83c91ca, 00m20s elapsed]

aws_instance.myec2: Still destroying... [id=i-014c7b457c83c91ca, 00m30s elapsed]

aws_instance.myec2: Destruction complete after 32s

Destroy complete! Resources: 2 destroyed.

2. Create 2 EC2 instances in Ohio and N.Virginia respectively

```
provider "aws" {
```

```
  region = "us-east-2"
```

```
}
```

```
resource "aws_instance" "myec2" {
```

```
  ami           = "ami-0741dc526e1106ae5" # Amazon Linux 2 AMI
```

```
  instance_type = "t3.micro"
```

```

tags = {

  Name = "Terraform-EC2"

}

}

provider "aws" {

  alias = "virginia"

  region = "us-east-1"

}

resource "aws_instance" "myec2_virginia" {

  provider      = aws.virginia

  ami           = "ami-0d0d444781b37ccdf" # Amazon Linux 2 AMI

  instance_type = "t3.micro"

  tags = {

    Name = "Terraform-EC2"

  }

}

```

Instances (1) [Info](#)

Last updated 1 minute ago

Connect

Instance state ▾

Actions ▾

Launch instances ▾

Saved filter sets

Choose filter set ▾

Find Instance by attribute or tag (case-sensitive)

< 1 > ⚙

☐	Name 🔗	Instance ID	Instance state ▾	Instance type ▾	Status check	Availability
☐	Terraform-EC2	i-0a30df65d1f8af1c5	Running 🔄 🔍	t3.micro	⌚ Initializing	us-east-1c

Instances (1) Info		Last updated 1 minute ago	Connect	Instance state ▼	Actions ▼	Launch instance
Saved filter sets Running ▼		Find Instance by attribute or tag (case-sensitive)				
Instance state = running X		Clear filters ▼				
		< 1				
☐	Name	Instance ID	Instance state	Instance type	Status check	Availability
☐	Terraform-EC2	i-017da3991f6b18b71	Running	t3.micro	3/3 checks passed	us-east-2

3. Rename Ohio's instance to 'hello-ohio' and Virginia's instance to 'hello-virginia'

PS C:\Users\jayan\Terraform\Terraform_practice> terraform state list

aws_instance.myec2

aws_instance.myec2_virginia

PS C:\Users\jayan\Terraform\Terraform_practice> terraform state mv aws_instance.myec2 aws_instance.hello-ohio

Move "aws_instance.myec2" to "aws_instance.hello-ohio"

Successfully moved 1 object(s).

PS C:\Users\jayan\Terraform\Terraform_practice> terraform state list

aws_instance.hello-ohio

aws_instance.myec2_virginia

PS C:\Users\jayan\Terraform\Terraform_practice> terraform state mv aws_instance.myec2_virginia aws_instance.hello-virginia

Move "aws_instance.myec2_virginia" to "aws_instance.hello-virginia"

Successfully moved 1 object(s).

PS C:\Users\jayan\Terraform\Terraform_practice> terraform state list

aws_instance.hello-ohio

aws_instance.hello-virginia

Assignment : 4

Tasks To Be Performed:

1. Destroy the previous deployments

```
PS C:\Users\jayan\Terraform\Terraform_practice> terraform destroy -auto-approve
```

```
aws_instance.hello-ohio: Refreshing state... [id=i-017da3991f6b18b71]
```

```
aws_instance.hello-virginia: Refreshing state... [id=i-0a30df65d1f8af1c5]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the

following symbols:

- destroy

Terraform will perform the following actions:

```
# aws_instance.hello-ohio will be destroyed
```

```
- resource "aws_instance" "hello-ohio" {  
  - ami                  = "ami-0741dc526e1106ae5" -> null  
  - arn                  = "arn:aws:ec2:us-east-2:931628309022:instance/i-017da3991f6b18b71" -> null  
  - associate_public_ip_address    = true -> null  
  - availability_zone              = "us-east-2c" -> null  
  - disable_api_stop               = false -> null  
  - disable_api_termination        = false -> null  
  - ebs_optimized                  = false -> null  
  - force_destroy                  = false -> null  
  - get_password_data              = false -> null  
  - hibernation                    = false -> null  
  - id                            = "i-017da3991f6b18b71" -> null  
  - instance_initiated_shutdown_behavior = "stop" -> null  
}
```

.

```
- tags_all                = {  
  - "Name" = "Terraform-EC2"  
}  
-> null  
  
- tenancy                  = "default" -> null  
  
- user_data_replace_on_change = false -> null  
  
- vpc_security_group_ids    = [  
  - "sg-0c68c12a93a123604",  
]  
-> null  
  
# (9 unchanged attributes hidden)  
  
- capacity_reservation_specification {  
  - capacity_reservation_preference = "open" -> null  
}  
  
- cpu_options {  
  - core_count      = 1 -> null  
  - threads_per_core = 2 -> null  
  # (2 unchanged attributes hidden)  
}  
  
- credit_specification {  
  - cpu_credits = "unlimited" -> null  
}  
  
- enclave_options {  
  - enabled = false -> null  
}
```

```
- maintenance_options {  
  
  - auto_recovery = "default" -> null  
  
}  
  
- metadata_options {  
  
  - http_endpoint      = "enabled" -> null  
  
  - http_protocol_ipv6  = "disabled" -> null  
  
  - http_put_response_hop_limit = 2 -> null  
  
  - http_tokens        = "required" -> null  
  
  - instance_metadata_tags = "disabled" -> null  
  
}  
  
- primary_network_interface {  
  
  - delete_on_termination = true -> null  
  
  - network_interface_id = "eni-0441ab8de8b3a2817" -> null  
  
}  
  
- private_dns_name_options {  
  
  - enable_resource_name_dns_a_record = false -> null  
  
  - enable_resource_name_dns_aaaa_record = false -> null  
  
  - hostname_type = "ip-name" -> null  
  
}  
  
- root_block_device {  
  
  - delete_on_termination = true -> null  
  
  - device_name = "/dev/xvda" -> null
```

```

- encrypted          = false -> null

- iops               = 3000 -> null

- tags               = {} -> null

- tags_all           = {} -> null

- throughput         = 125 -> null

- volume_id          = "vol-0392cbaa73d15c562" -> null

- volume_size        = 8 -> null

- volume_type        = "gp3" -> null

# (1 unchanged attribute hidden)

}

}

# aws_instance.hello-virginia will be destroyed

- resource "aws_instance" "hello-virginia" {

- ami                = "ami-0d0d444781b37ccdf" -> null

- arn                 = "arn:aws:ec2:us-east-1:931628309022:instance/i-0a30df65d1f8af1c5" -> null

- associate_public_ip_address    = true -> null

- availability_zone      = "us-east-1c" -> null

- disable_api_stop       = false -> null

- disable_api_termination = false -> null

- ebs_optimized           = false -> null

- force_destroy           = false -> null

- get_password_data       = false -> null

- hibernation              = false -> null

- id                      = "i-0a30df65d1f8af1c5" -> null

- instance_initiated_shutdown_behavior = "stop" -> null

- instance_state          = "running" -> null

```

```
- instance_type            = "t3.micro" -> null

- ipv6_address_count       = 0 -> null

- ipv6_addresses           = [] -> null

- monitoring               = false -> null

- placement_partition_number = 0 -> null

- primary_network_interface_id = "eni-03fc36f0a695807c7" -> null

- private_dns              = "ip-172-31-25-240.ec2.internal" -> null

- private_ip               = "172.31.25.240" -> null

- public_dns               = "ec2-18-209-49-144.compute-1.amazonaws.com" -> null

- public_ip                = "18.209.49.144" -> null

- region                   = "us-east-1" -> null

- secondary_private_ips     = [] -> null

- security_groups           = [

  - "default",

] -> null

- source_dest_check         = true -> null

- subnet_id                = "subnet-004090f6e31b637f7" -> null

- tags                     = {

  - "Name" = "Terraform-EC2"

} -> null

- tags_all                 = {

  - "Name" = "Terraform-EC2"

} -> null

- tenancy                  = "default" -> null

- user_data_replace_on_change = false -> null

- vpc_security_group_ids    = [

  - "sg-0c0572f218f694386",
```


] -> null

(9 unchanged attributes hidden)

- capacity_reservation_specification {

- capacity_reservation_preference = "open" -> null

}

- cpu_options {

- core_count = 1 -> null

- threads_per_core = 2 -> null

(2 unchanged attributes hidden)

}

- credit_specification {

- cpu_credits = "unlimited" -> null

}

- enclave_options {

- enabled = false -> null

}

- maintenance_options {

- auto_recovery = "default" -> null

}

- metadata_options {

- http_endpoint = "enabled" -> null

```
- http_protocol_ipv6      = "disabled" -> null

- http_put_response_hop_limit = 2 -> null

- http_tokens              = "required" -> null

- instance_metadata_tags   = "disabled" -> null

}

- primary_network_interface {

  - delete_on_termination = true -> null

  - network_interface_id = "eni-03fc36f0a695807c7" -> null

}

- private_dns_name_options {

  - enable_resource_name_dns_a_record  = false -> null

  - enable_resource_name_dns_aaaa_record = false -> null

  - hostname_type                      = "ip-name" -> null

}

- root_block_device {

  - delete_on_termination = true -> null

  - device_name           = "/dev/xvda" -> null

  - encrypted             = false -> null

  - iops                  = 3000 -> null

  - tags                  = {} -> null

  - tags_all              = {} -> null

  - throughput            = 125 -> null

  - volume_id             = "vol-0b49983e9f75121f5" -> null

  - volume_size           = 8 -> null
```

```
- volume_type      = "gp3" -> null
```

```
# (1 unchanged attribute hidden)
```

```
}
```

```
}
```

Plan: 0 to add, 0 to change, 2 to destroy.

aws_instance.hello-ohio: Destroying... [id=i-017da3991f6b18b71]

aws_instance.hello-virginia: Destroying... [id=i-0a30df65d1f8af1c5]

aws_instance.hello-ohio: Still destroying... [id=i-017da3991f6b18b71, 00m10s elapsed]

aws_instance.hello-virginia: Still destroying... [id=i-0a30df65d1f8af1c5, 00m10s elapsed]

aws_instance.hello-ohio: Still destroying... [id=i-017da3991f6b18b71, 00m20s elapsed]

aws_instance.hello-virginia: Still destroying... [id=i-0a30df65d1f8af1c5, 00m20s elapsed]

aws_instance.hello-virginia: Destruction complete after 22s

aws_instance.hello-ohio: Still destroying... [id=i-017da3991f6b18b71, 00m30s elapsed]

aws_instance.hello-ohio: Destruction complete after 33s

Destroy complete! Resources: 2 destroyed.

PS C:\Users\jayan\Terraform\Terraform_practice>

2. Create a VPC with the required components using Terraform

3. Deploy an EC2 instance inside the VPC

1. **Create VPC**
2. **Create public and private subnets**
3. **Create IGW and attach to VPC**
4. **Create route table**
5. **Associate route table to public subnet and add a route to IGW**
6. **Create SG with ssh and HTTP rule**
7. **Create EC2 instance under defined VPC and public subnet**

PS C:\Users\jayan\Terraform\Terraform_practice\terraform_vpc> terraform apply -auto-approve

aws_vpc.main: Refreshing state... [id=vpc-03c82286b22e3f973]

aws_internet_gateway.my_igw: Refreshing state... [id=igw-01645dcdef08066bf]

aws_subnet.private_sb: Refreshing state... [id=subnet-0152a2b6e3cef9f06]

aws_subnet.public_sb: Refreshing state... [id=subnet-0b112a29706cffaf0]

aws_security_group.my_sg: Refreshing state... [id=sg-09e3a590b8372aadd]

aws_route_table.public_RT: Refreshing state... [id=rtb-00b986f3ae9aa4ad9]

aws_vpc_security_group_egress_rule.allow_all_traffic_ipv6: Refreshing state... [id=sgr-0721836483e607067]

aws_vpc_security_group_egress_rule.allow_all_traffic_ipv4: Refreshing state... [id=sgr-031000c5697024e92]

aws_vpc_security_group_ingress_rule.inbound_ssh: Refreshing state... [id=sgr-07876b772b53a3de0]

aws_vpc_security_group_ingress_rule.inbound_http: Refreshing state... [id=sgr-074d6c468c6b65cb6]

aws_instance.my_tf_instance: Refreshing state... [id=i-08e0ca650457a3af6]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the

following symbols:

+ create

-/+ destroy and then create replacement

Terraform will perform the following actions:

aws_instance.my_tf_instance must be replaced

-/+ resource "aws_instance" "my_tf_instance" {

~ arn = "arn:aws:ec2:ap-south-1:931628309022:instance/i-08e0ca650457a3af6" -> (known after apply)

~ associate_public_ip_address = false -> true # forces replacement

~ availability_zone = "ap-south-1c" -> (known after apply)

~ disable_api_stop = false -> (known after apply)

~ disable_api_termination = false -> (known after apply)

~ ebs_optimized = false -> (known after apply)

+ enable_primary_ipv6 = (known after apply)

- hibernation = false -> null

+ host_id = (known after apply)

+ host_resource_group_arn = (known after apply)

+ iam_instance_profile = (known after apply)

~ id = "i-08e0ca650457a3af6" -> (known after apply)

~ instance_initiated_shutdown_behavior = "stop" -> (known after apply)

+ instance_lifecycle = (known after apply)

~ instance_state = "running" -> (known after apply)

~ ipv6_address_count = 0 -> (known after apply)

~ ipv6_addresses = [] -> (known after apply)

+ key_name = (known after apply)

~ monitoring = false -> (known after apply)

+ outpost_arn = (known after apply)

+ password_data = (known after apply)

+ placement_group = (known after apply)

+ placement_group_id = (known after apply)

~ placement_partition_number = 0 -> (known after apply)

~ primary_network_interface_id = "eni-0b388a9ced1947f5d" -> (known after apply)

~ private_dns = "ip-10-0-1-193.ap-south-1.compute.internal" -> (known after apply)

~ private_ip = "10.0.1.193" -> (known after apply)

+ public_dns = (known after apply)

+ public_ip = (known after apply)

~ secondary_private_ips = [] -> (known after apply)

~ security_groups = [] -> (known after apply)

+ spot_instance_request_id = (known after apply)

- tags = {} -> null

~ tags_all = {} -> (known after apply)

~ tenancy = "default" -> (known after apply)

+ user_data_base64 = (known after apply)

(9 unchanged attributes hidden)

~ capacity_reservation_specification (known after apply)

- capacity_reservation_specification {

- capacity_reservation_preference = "open" -> null

}

~ cpu_options (known after apply)

- cpu_options {

- core_count = 1 -> null

- threads_per_core = 2 -> null

(2 unchanged attributes hidden)

}

- credit_specification {

- cpu_credits = "unlimited" -> null

}

~ ebs_block_device (known after apply)

~ enclave_options (known after apply)

- enclave_options {

- enabled = false -> null

}

~ ephemeral_block_device (known after apply)

~ instance_market_options (known after apply)

~ maintenance_options (known after apply)

```
- maintenance_options {  
  - auto_recovery = "default" -> null  
}
```

~ metadata_options (known after apply)

```
- metadata_options {  
  - http_endpoint      = "enabled" -> null  
  - http_protocol_ipv6 = "disabled" -> null  
  - http_put_response_hop_limit = 2 -> null  
  - http_tokens        = "required" -> null  
  - instance_metadata_tags = "disabled" -> null  
}
```

~ network_interface (known after apply)

~ primary_network_interface (known after apply)

```
- primary_network_interface {  
  - delete_on_termination = true -> null  
  - network_interface_id = "eni-0b388a9ced1947f5d" -> null  
}
```

~ private_dns_name_options (known after apply)

```
- private_dns_name_options {

  - enable_resource_name_dns_a_record  = false -> null

  - enable_resource_name_dns_aaaa_record = false -> null

  - hostname_type                      = "ip-name" -> null

}
```

~ root_block_device (known after apply)

```
- root_block_device {

  - delete_on_termination = true -> null

  - device_name           = "/dev/xvda" -> null

  - encrypted             = false -> null

  - iops                  = 3000 -> null

  - tags                  = {} -> null

  - tags_all              = {} -> null

  - throughput            = 125 -> null

  - volume_id             = "vol-0d1177a77b0fe832c" -> null

  - volume_size           = 8 -> null

  - volume_type            = "gp3" -> null

  # (1 unchanged attribute hidden)

}
```

~ secondary_network_interface (known after apply)

```
}
```

aws_route_table.public_RT is tainted, so must be replaced

```
-/+ resource "aws_route_table" "public_RT" {
```

```
  ~ arn          = "arn:aws:ec2:ap-south-1:931628309022:route-table/rtb-00b986f3ae9aa4ad9" -> (known after
apply)
```



```

~ id          = "rtb-00b986f3ae9aa4ad9" -> (known after apply)

~ owner_id    = "931628309022" -> (known after apply)

~ propagating_vgws = [] -> (known after apply)

~ route       = [

  + {

    + cidr_block      = "0.0.0.0/0"

    + gateway_id      = "igw-01645dcdef08066bf"

    # (12 unchanged attributes hidden)

  },

]

tags          = {

  "Name" = "Route_table_public"

}

# (3 unchanged attributes hidden)

}

```

aws_route_table_association.public_RT_association will be created

```

+ resource "aws_route_table_association" "public_RT_association" {

  + id          = (known after apply)

  + region      = "ap-south-1"

  + route_table_id = (known after apply)

  + subnet_id    = "subnet-0b112a29706cfffaf0"

}

```

Plan: 3 to add, 0 to change, 2 to destroy.

aws_route_table.public_RT: Destroying... [id=rtb-00b986f3ae9aa4ad9]

aws_instance.my_tf_instance: Destroying... [id=i-08e0ca650457a3af6]

aws_route_table.public_RT: Destruction complete after 1s

aws_route_table.public_RT: Creating...

aws_route_table.public_RT: Creation complete after 2s [id=rtb-Of3d3cc3ea8ce8cb2]

aws_route_table_association.public_RT_association: Creating...

aws_route_table_association.public_RT_association: Creation complete after 0s [id=rtbassoc-0bc13bdec86240786]

aws_instance.my_tf_instance: Still destroying... [id=i-08e0ca650457a3af6, 00m10s elapsed]

aws_instance.my_tf_instance: Still destroying... [id=i-08e0ca650457a3af6, 00m20s elapsed]

aws_instance.my_tf_instance: Still destroying... [id=i-08e0ca650457a3af6, 00m30s elapsed]

aws_instance.my_tf_instance: Destruction complete after 31s

aws_instance.my_tf_instance: Creating...

aws_instance.my_tf_instance: Still creating... [00m10s elapsed]

aws_instance.my_tf_instance: Creation complete after 13s [id=i-0645e7f97a7490fc7]

Apply complete! Resources: 3 added, 0 changed, 2 destroyed.

PS C:\Users\jayan\Terraform\Terraform_practice\terraform_vpc> terraform show

aws_instance.my_tf_instance:

```
resource "aws_instance" "my_tf_instance" {  
  
    ami                = "ami-0e38835daf6b8a2b9"  
  
    arn                = "arn:aws:ec2:ap-south-1:931628309022:instance/i-0645e7f97a7490fc7"  
  
    associate_public_ip_address    = true  
  
    availability_zone    = "ap-south-1c"  
  
    disable_api_stop      = false  
  
    disable_api_termination    = false  
  
    ebs_optimized          = false  
  
    force_destroy          = false  
  
    get_password_data      = false  
  
    hibernation            = false
```

host_id = null

iam_instance_profile = null

id = "i-0645e7f97a7490fc7"

instance_initiated_shutdown_behavior = "stop"

instance_lifecycle = null

instance_state = "running"

instance_type = "t3.micro"

ipv6_address_count = 0

ipv6_addresses = []

key_name = null

monitoring = false

outpost_arn = null

password_data = null

placement_group = null

placement_group_id = null

placement_partition_number = 0

primary_network_interface_id = "eni-023885fd18bcd4d75"

private_dns = "ip-10-0-1-42.ap-south-1.compute.internal"

private_ip = "10.0.1.42"

public_dns = null

public_ip = "13.204.72.108"

region = "ap-south-1"

secondary_private_ips = []

security_groups = []

source_dest_check = true

spot_instance_request_id = null

subnet_id = "subnet-0b112a29706cffa0"

```
tags_all          = {}

tenancy           = "default"

user_data_replace_on_change = false

vpc_security_group_ids = [

    "sg-09e3a590b8372aadd",

]


capacity_reservation_specification {

    capacity_reservation_preference = "open"

}


cpu_options {

    amd_sev_snp      = null

    core_count       = 1

    nested_virtualization = null

    threads_per_core  = 2

}


credit_specification {

    cpu_credits = "unlimited"

}


enclave_options {

    enabled = false

}


maintenance_options {
```

```
    auto_recovery = "default"

}

metadata_options {

    http_endpoint      = "enabled"

    http_protocol_ipv6  = "disabled"

    http_put_response_hop_limit = 2

    http_tokens         = "required"

    instance_metadata_tags = "disabled"

}

primary_network_interface {

    delete_on_termination = true

    network_interface_id = "eni-023885fd18bcd4d75"

}

private_dns_name_options {

    enable_resource_name_dns_a_record  = false

    enable_resource_name_dns_aaaa_record = false

    hostname_type                      = "ip-name"

}

root_block_device {

    delete_on_termination = true

    device_name            = "/dev/xvda"

    encrypted              = false

    iops                   = 3000

}
```

```

    kms_key_id      = null

    tags            = {}

    tags_all        = {}

    throughput      = 125

    volume_id       = "vol-008ae105f962a650b"

    volume_size     = 8

    volume_type     = "gp3"

  }

}

# aws_internet_gateway.my_igw:

resource "aws_internet_gateway" "my_igw" {

  arn      = "arn:aws:ec2:ap-south-1:931628309022:internet-gateway/igw-01645dcdef08066bf"

  id       = "igw-01645dcdef08066bf"

  owner_id = "931628309022"

  region   = "ap-south-1"

  tags     = {

    "Name" = "main"

  }

  tags_all = {

    "Name" = "main"

  }

  vpc_id   = "vpc-03c82286b22e3f973"

}

# aws_route_table.public_RT:

resource "aws_route_table" "public_RT" {

```

arn = "arn:aws:ec2:ap-south-1:931628309022:route-table/rtb-Of3d3cc3ea8ce8cb2"

id = "rtb-Of3d3cc3ea8ce8cb2"

owner_id = "931628309022"

propagating_vgws = []

region = "ap-south-1"

route = [

{

carrier_gateway_id = null

cidr_block = "0.0.0.0/0"

core_network_arn = null

destination_prefix_list_id = null

egress_only_gateway_id = null

gateway_id = "igw-01645dcdef08066bf"

ipv6_cidr_block = null

local_gateway_id = null

nat_gateway_id = null

network_interface_id = null

odb_network_arn = null

transit_gateway_id = null

vpc_endpoint_id = null

vpc_peering_connection_id = null

},

]

tags = {

"Name" = "Route_table_public"

}

tags_all = {

```

    "Name" = "Route_table_public"

}

vpc_id      = "vpc-03c82286b22e3f973"

}

# aws_route_table_association.public_RT_association:

resource "aws_route_table_association" "public_RT_association" {

    gateway_id    = null

    id            = "rtbassoc-0bc13bdec86240786"

    region        = "ap-south-1"

    route_table_id = "rtb-0f3d3cc3ea8ce8cb2"

    subnet_id     = "subnet-0b112a29706cfffaf0"

}

# aws_security_group.my_sg:

resource "aws_security_group" "my_sg" {

    arn          = "arn:aws:ec2:ap-south-1:931628309022:security-group/sg-09e3a590b8372aadd"

    description   = "Allow TLS inbound traffic and all outbound traffic"

    egress        = [

        {

            cidr_blocks = [

                "0.0.0.0/0",

            ]

            description = null

            from_port   = 0

            ipv6_cidr_blocks = [

                ":::/0",

```



```
]

prefix_list_ids = []

protocol      = "-1"

security_groups = []

self         = false

to_port      = 0

},

]

id           = "sg-09e3a590b8372aadd"

ingress      = [

{

  cidr_blocks = [

    "0.0.0.0/0",

  ]

  description = null

  from_port   = 22

  ipv6_cidr_blocks = []

  prefix_list_ids = []

  protocol    = "tcp"

  security_groups = []

  self        = false

  to_port     = 22

},

{

  cidr_blocks = [

    "0.0.0.0/0",

  ]

}
```

```

        description    = null

        from_port      = 80

        ipv6_cidr_blocks = []

        prefix_list_ids = []

        protocol        = "tcp"

        security_groups = []

        self            = false

        to_port         = 80

    },

]

name            = "my_sg"

name_prefix     = null

owner_id        = "931628309022"

region         = "ap-south-1"

revoke_rules_on_delete = false

tags            = {

    "Name" = "my_sg"

}

tags_all        = {

    "Name" = "my_sg"

}

vpc_id          = "vpc-03c82286b22e3f973"

}

# aws_subnet.private_sb:

resource "aws_subnet" "private_sb" {

    arn                = "arn:aws:ec2:ap-south-1:931628309022:subnet/subnet-0152a2b6e3cef9f06"

```



```
# aws_subnet.public_sb:
```

```
resource "aws_subnet" "public_sb" {
```

```
    arn                        = "arn:aws:ec2:ap-south-1:931628309022:subnet/subnet-0b112a29706cfffaf0"
```

```
    assign_ipv6_address_on_creation    = false
```

```
    availability_zone                = "ap-south-1c"
```

```
    availability_zone_id              = "aps1-az2"
```

```
    cidr_block                       = "10.0.1.0/24"
```

```
    customer_owned_ipv4_pool          = null
```

```
    enable_dns64                      = false
```

```
    enable_lni_at_device_index        = 0
```

```
    enable_resource_name_dns_a_record_on_launch = false
```

```
    enable_resource_name_dns_aaaa_record_on_launch = false
```

```
    id                                = "subnet-0b112a29706cfffaf0"
```

```
    ipv6_cidr_block                   = null
```

```
    ipv6_cidr_block_association_id    = null
```

```
    ipv6_native                       = false
```

```
    map_customer_owned_ip_on_launch    = false
```

```
    map_public_ip_on_launch            = false
```

```
    outpost_arn                       = null
```

```
    owner_id                          = "931628309022"
```

```
    private_dns_hostname_type_on_launch = "ip-name"
```

```
    region                            = "ap-south-1"
```

```
    tags                              = {
```

```
        "Name" = "public_subnet"
```

```
    }
```

```
    tags_all                          = {
```

```

    "Name" = "public_subnet"

}

vpc_id = "vpc-03c82286b22e3f973"

}

# aws_vpc.main:

resource "aws_vpc" "main" {

    arn = "arn:aws:ec2:ap-south-1:931628309022:vpc/vpc-03c82286b22e3f973"

    assign_generated_ipv6_cidr_block = false

    cidr_block = "10.0.0.0/16"

    default_network_acl_id = "acl-0e1c4ba24f5d4b847"

    default_route_table_id = "rtb-017b4f84d9239fd26"

    default_security_group_id = "sg-0eb4b19336ec47d9f"

    dhcp_options_id = "dopt-04b7a6eba4c9859c3"

    enable_dns_hostnames = false

    enable_dns_support = true

    enable_network_address_usage_metrics = false

    id = "vpc-03c82286b22e3f973"

    instance_tenancy = "default"

    ipv6_association_id = null

    ipv6_cidr_block = null

    ipv6_cidr_block_network_border_group = null

    ipv6_ipam_pool_id = null

    ipv6_netmask_length = 0

    main_route_table_id = "rtb-017b4f84d9239fd26"

    owner_id = "931628309022"

    region = "ap-south-1"

```

```

tags                = {}

tags_all            = {}

}

# aws_vpc_security_group_egress_rule.allow_all_traffic_ipv4:

resource "aws_vpc_security_group_egress_rule" "allow_all_traffic_ipv4" {

  arn                = "arn:aws:ec2:ap-south-1:931628309022:security-group-rule/sgr-031000c5697024e92"

  cidr_ipv4          = "0.0.0.0/0"

  id                 = "sgr-031000c5697024e92"

  ip_protocol        = "-1"

  region             = "ap-south-1"

  security_group_id   = "sg-09e3a590b8372aadd"

  security_group_rule_id = "sgr-031000c5697024e92"

  tags_all           = {}

}

# aws_vpc_security_group_egress_rule.allow_all_traffic_ipv6:

resource "aws_vpc_security_group_egress_rule" "allow_all_traffic_ipv6" {

  arn                = "arn:aws:ec2:ap-south-1:931628309022:security-group-rule/sgr-0721836483e607067"

  cidr_ipv6          = "::/0"

  id                 = "sgr-0721836483e607067"

  ip_protocol        = "-1"

  region             = "ap-south-1"

  security_group_id   = "sg-09e3a590b8372aadd"

  security_group_rule_id = "sgr-0721836483e607067"

  tags_all           = {}

}

```

```
# aws_vpc_security_group_ingress_rule.inbound_http:
```

```
resource "aws_vpc_security_group_ingress_rule" "inbound_http" {
```

```
    arn          = "arn:aws:ec2:ap-south-1:931628309022:security-group-rule/sgr-074d6c468c6b65cb6"
```

```
    cidr_ipv4     = "0.0.0.0/0"
```

```
    from_port     = 80
```

```
    id           = "sgr-074d6c468c6b65cb6"
```

```
    ip_protocol   = "tcp"
```

```
    region       = "ap-south-1"
```

```
    security_group_id = "sg-09e3a590b8372aadd"
```

```
    security_group_rule_id = "sgr-074d6c468c6b65cb6"
```

```
    tags_all      = {}
```

```
    to_port       = 80
```

```
}
```

```
# aws_vpc_security_group_ingress_rule.inbound_ssh:
```

```
resource "aws_vpc_security_group_ingress_rule" "inbound_ssh" {
```

```
    arn          = "arn:aws:ec2:ap-south-1:931628309022:security-group-rule/sgr-07876b772b53a3de0"
```

```
    cidr_ipv4     = "0.0.0.0/0"
```

```
    from_port     = 22
```

```
    id           = "sgr-07876b772b53a3de0"
```

```
    ip_protocol   = "tcp"
```

```
    region       = "ap-south-1"
```

```
    security_group_id = "sg-09e3a590b8372aadd"
```

```
    security_group_rule_id = "sgr-07876b772b53a3de0"
```

```
    tags_all      = {}
```

```
    to_port       = 22
```

```
}
```

```
PS C:\Users\jayan\Terraform\Terraform_practice\terraform_vpc>
```

```
main.tf
```

```
# provider
```

```
provider "aws" {
```

```
    region = "ap-south-1"
```

```
}
```

```
# Create VPC
```

```
resource "aws_vpc" "main" {
```

```
    cidr_block = "10.0.0.0/16"
```

```
}
```

```
# Create public and private subnets
```

```
resource "aws_subnet" "public_sb" {
```

```
    vpc_id      = aws_vpc.main.id
```

```
    cidr_block = "10.0.1.0/24"
```

```
    tags = {
```

```
        Name = "public_subnet"
```

```
    }
```

```
}
```

```
resource "aws_subnet" "private_sb" {
```

```
    vpc_id      = aws_vpc.main.id
```

```
    cidr_block = "10.0.2.0/24"
```

```
    tags = {
```

```
        Name = "private_subnet"
```



```
}
```

```
}
```

Create IGW and attach to VPC

```
resource "aws_internet_gateway" "my_igw" {
```

```
  vpc_id = aws_vpc.main.id
```

```
  tags = {
```

```
    Name = "main"
```

```
  }
```

```
}
```

Create route table and add a route to IGW

```
resource "aws_route_table" "public_RT" {
```

```
  vpc_id = aws_vpc.main.id
```

```
  route {
```

```
    cidr_block = "0.0.0.0/0"
```

```
    gateway_id = aws_internet_gateway.my_igw.id
```

```
  }
```

```
  tags = {
```

```
    Name = "Route_table_public"
```

```
  }
```

```
}
```

Associate route table to public subnet

```
resource "aws_route_table_association" "public_RT_association" {
```

```
  subnet_id    = aws_subnet.public_sb.id
```

```
  route_table_id = aws_route_table.public_RT.id
```

```
}
```

```
# Create SG with ssh and HTTP rule
```

```
resource "aws_security_group" "my_sg" {
```

```
    name      = "my_sg"
```

```
    description = "Allow TLS inbound traffic and all outbound traffic"
```

```
    vpc_id     = aws_vpc.main.id
```

```
    tags = {
```

```
        Name = "my_sg"
```

```
    }
```

```
}
```

```
resource "aws_vpc_security_group_ingress_rule" "inbound_ssh" {
```

```
    security_group_id = aws_security_group.my_sg.id
```

```
    cidr_ipv4      = "0.0.0.0/0"
```

```
    from_port      = 22
```

```
    ip_protocol     = "tcp"
```

```
    to_port        = 22
```

```
}
```

```
resource "aws_vpc_security_group_ingress_rule" "inbound_http" {
```

```
    security_group_id = aws_security_group.my_sg.id
```

```
    cidr_ipv4      = "0.0.0.0/0"
```

```
    from_port      = 80
```

```
    ip_protocol     = "tcp"
```

```
    to_port        = 80
```

```
}
```

```

resource "aws_vpc_security_group_egress_rule" "allow_all_traffic_ipv4" {

  security_group_id = aws_security_group.my_sg.id

  cidr_ipv4      = "0.0.0.0/0"

  ip_protocol    = "-1" # semantically equivalent to all ports

}

```

```

resource "aws_vpc_security_group_egress_rule" "allow_all_traffic_ipv6" {

  security_group_id = aws_security_group.my_sg.id

  cidr_ipv6      = "::/0"

  ip_protocol    = "-1" # semantically equivalent to all ports

}

```

Create EC2 instance under defined VPC and public subnet

```

resource "aws_instance" "my_tf_instance" {

  ami          = "ami-0e38835daf6b8a2b9"

  instance_type    = "t3.micro"

  subnet_id       = aws_subnet.public_sb.id

  vpc_security_group_ids = [aws_security_group.my_sg.id]

  associate_public_ip_address = true

}

```

Assignment :5

Tasks To Be Performed:

1. Destroy the previous deployments

PS C:\Users\jayan\Terraform\Terraform_practice\terraform_vpc> terraform destroy -auto-approve

aws_vpc.main: Refreshing state... [id=vpc-03c82286b22e3f973]

aws_internet_gateway.my_igw: Refreshing state... [id=igw-01645dcdef08066bf]

aws_subnet.public_sb: Refreshing state... [id=subnet-0b112a29706cffaf0]

aws_subnet.private_sb: Refreshing state... [id=subnet-0152a2b6e3cef9f06]

aws_security_group.my_sg: Refreshing state... [id=sg-09e3a590b8372aadd]

aws_route_table.public_RT: Refreshing state... [id=rtb-0f3d3cc3ea8ce8cb2]

aws_vpc_security_group_egress_rule.allow_all_traffic_ipv4: Refreshing state... [id=sgr-031000c5697024e92]

aws_vpc_security_group_egress_rule.allow_all_traffic_ipv6: Refreshing state... [id=sgr-0721836483e607067]

aws_vpc_security_group_ingress_rule.inbound_ssh: Refreshing state... [id=sgr-07876b772b53a3de0]

aws_vpc_security_group_ingress_rule.inbound_http: Refreshing state... [id=sgr-074d6c468c6b65cb6]

aws_instance.my_tf_instance: Refreshing state... [id=i-0645e7f97a7490fc7]

aws_route_table_association.public_RT_association: Refreshing state... [id=rtbassoc-0bc13bdec86240786]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the

following symbols:

- destroy

Terraform will perform the following actions:

aws_instance.my_tf_instance will be destroyed

- resource "aws_instance" "my_tf_instance" {

- ami = "ami-0e38835daf6b8a2b9" -> null

- arn = "arn:aws:ec2:ap-south-1:931628309022:instance/i-0645e7f97a7490fc7" -> null

- associate_public_ip_address = true -> null

- availability_zone = "ap-south-1c" -> null

- disable_api_stop = false -> null

- disable_api_termination = false -> null

- ebs_optimized = false -> null

- force_destroy = false -> null

- get_password_data = false -> null

- hibernation = false -> null

- id = "i-0645e7f97a7490fc7" -> null

- instance_initiated_shutdown_behavior = "stop" -> null

- instance_state = "running" -> null

- instance_type = "t3.micro" -> null

- ipv6_address_count = 0 -> null

- ipv6_addresses = [] -> null

- monitoring = false -> null

- placement_partition_number = 0 -> null

- primary_network_interface_id = "eni-023885fd18bcd4d75" -> null

- private_dns = "ip-10-0-1-42.ap-south-1.compute.internal" -> null

- private_ip = "10.0.1.42" -> null

- public_ip = "13.204.72.108" -> null

- region = "ap-south-1" -> null

- secondary_private_ips = [] -> null

- security_groups = [] -> null

- source_dest_check = true -> null

- subnet_id = "subnet-0b112a29706cfffaf0" -> null

- tags = {} -> null

- tags_all = {} -> null

- tenancy = "default" -> null

- user_data_replace_on_change = false -> null

- vpc_security_group_ids = [
- "sg-09e3a590b8372aadd",
] -> null

(10 unchanged attributes hidden)

```
- capacity_reservation_specification {  
    - capacity_reservation_preference = "open" -> null  
}
```

```
- cpu_options {  
    - core_count      = 1 -> null  
    - threads_per_core = 2 -> null  
    # (2 unchanged attributes hidden)  
}
```

```
- credit_specification {  
    - cpu_credits = "unlimited" -> null  
}
```

```
- enclave_options {  
    - enabled = false -> null  
}
```

```
- maintenance_options {  
    - auto_recovery = "default" -> null  
}
```

```
- metadata_options {  
    - http_endpoint      = "enabled" -> null  
    - http_protocol_ipv6 = "disabled" -> null  
    - http_put_response_hop_limit = 2 -> null  
    - http_tokens        = "required" -> null  
}
```

```
- instance_metadata_tags    = "disabled" -> null

}

- primary_network_interface {

  - delete_on_termination = true -> null

  - network_interface_id = "eni-023885fd18bcd4d75" -> null

}

- private_dns_name_options {

  - enable_resource_name_dns_a_record    = false -> null

  - enable_resource_name_dns_aaaa_record = false -> null

  - hostname_type                        = "ip-name" -> null

}

- root_block_device {

  - delete_on_termination = true -> null

  - device_name           = "/dev/xvda" -> null

  - encrypted             = false -> null

  - iops                  = 3000 -> null

  - tags                  = {} -> null

  - tags_all              = {} -> null

  - throughput            = 125 -> null

  - volume_id             = "vol-008ae105f962a650b" -> null

  - volume_size           = 8 -> null

  - volume_type           = "gp3" -> null

  # (1 unchanged attribute hidden)

}
```

```
}
```

```
# aws_internet_gateway.my_igw will be destroyed
```

```
- resource "aws_internet_gateway" "my_igw" {  
  
  - arn    = "arn:aws:ec2:ap-south-1:931628309022:internet-gateway/igw-01645dcdef08066bf" -> null  
  
  - id     = "igw-01645dcdef08066bf" -> null  
  
  - owner_id = "931628309022" -> null  
  
  - region  = "ap-south-1" -> null  
  
  - tags    = {  
  
    - "Name" = "main"  
  
  } -> null  
  
  - tags_all = {  
  
    - "Name" = "main"  
  
  } -> null  
  
  - vpc_id  = "vpc-03c82286b22e3f973" -> null  
  
}
```

```
# aws_route_table.public_RT will be destroyed
```

```
- resource "aws_route_table" "public_RT" {  
  
  - arn      = "arn:aws:ec2:ap-south-1:931628309022:route-table/rtb-0f3d3cc3ea8ce8cb2" -> null  
  
  - id       = "rtb-0f3d3cc3ea8ce8cb2" -> null  
  
  - owner_id = "931628309022" -> null  
  
  - propagating_vgws = [] -> null  
  
  - region      = "ap-south-1" -> null  
  
  - route       = [  
  
    - {  
  
      - cidr_block      = "0.0.0.0/0"
```



```

- gateway_id      = "igw-01645dcdef08066bf"

# (12 unchanged attributes hidden)

},

] -> null

- tags            = {

  - "Name" = "Route_table_public"

} -> null

- tags_all        = {

  - "Name" = "Route_table_public"

} -> null

- vpc_id          = "vpc-03c82286b22e3f973" -> null

}

# aws_route_table_association.public_RT_association will be destroyed

- resource "aws_route_table_association" "public_RT_association" {

  - id            = "rtbassoc-0bc13bdec86240786" -> null

  - region        = "ap-south-1" -> null

  - route_table_id = "rtb-0f3d3cc3ea8ce8cb2" -> null

  - subnet_id     = "subnet-0b112a29706cffaf0" -> null

  # (1 unchanged attribute hidden)

}

# aws_security_group.my_sg will be destroyed

- resource "aws_security_group" "my_sg" {

  - arn           = "arn:aws:ec2:ap-south-1:931628309022:security-group/sg-09e3a590b8372aadd" -> null

  - description   = "Allow TLS inbound traffic and all outbound traffic" -> null

  - egress        = [

```

```
- {  
  
  - cidr_blocks    = [  
  
    - "0.0.0.0/0",  
  
  ]  
  
  - from_port      = 0  
  
  - ipv6_cidr_blocks = [  
  
    - "::/0",  
  
  ]  
  
  - prefix_list_ids = []  
  
  - protocol        = "-1"  
  
  - security_groups = []  
  
  - self            = false  
  
  - to_port         = 0  
  
  # (1 unchanged attribute hidden)  
  
},  
  
] -> null  
  
- id              = "sg-09e3a590b8372aadd" -> null  
  
- ingress         = [  
  
  - {  
  
    - cidr_blocks    = [  
  
      - "0.0.0.0/0",  
  
    ]  
  
    - from_port      = 22  
  
    - ipv6_cidr_blocks = []  
  
    - prefix_list_ids = []  
  
    - protocol        = "tcp"  
  
    - security_groups = []
```

```

- self      = false

- to_port   = 22

# (1 unchanged attribute hidden)

},

- {

- cidr_blocks = [

- "0.0.0.0/0",

]

- from_port   = 80

- ipv6_cidr_blocks = []

- prefix_list_ids = []

- protocol    = "tcp"

- security_groups = []

- self      = false

- to_port   = 80

# (1 unchanged attribute hidden)

},

] -> null

- name      = "my_sg" -> null

- owner_id  = "931628309022" -> null

- region    = "ap-south-1" -> null

- revoke_rules_on_delete = false -> null

- tags      = {

- "Name" = "my_sg"

} -> null

- tags_all  = {

- "Name" = "my_sg"

```

```
} -> null
```

```
- vpc_id = "vpc-03c82286b22e3f973" -> null
```

```
# (1 unchanged attribute hidden)
```

```
}
```

```
# aws_subnet.private_sb will be destroyed
```

```
- resource "aws_subnet" "private_sb" {
```

```
  - arn = "arn:aws:ec2:ap-south-1:931628309022:subnet/subnet-0152a2b6e3cef9f06" -> null
```

```
  - assign_ipv6_address_on_creation = false -> null
```

```
  - availability_zone = "ap-south-1c" -> null
```

```
  - availability_zone_id = "aps1-az2" -> null
```

```
  - cidr_block = "10.0.2.0/24" -> null
```

```
  - enable_dns64 = false -> null
```

```
  - enable_lni_at_device_index = 0 -> null
```

```
  - enable_resource_name_dns_a_record_on_launch = false -> null
```

```
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
```

```
  - id = "subnet-0152a2b6e3cef9f06" -> null
```

```
  - ipv6_native = false -> null
```

```
  - map_customer_owned_ip_on_launch = false -> null
```

```
  - map_public_ip_on_launch = false -> null
```

```
  - owner_id = "931628309022" -> null
```

```
  - private_dns_hostname_type_on_launch = "ip-name" -> null
```

```
  - region = "ap-south-1" -> null
```

```
  - tags = {
```

```
    - "Name" = "private_subnet"
```

```
  } -> null
```

```
  - tags_all = {
```

```

- "Name" = "private_subnet"

} -> null

- vpc_id = "vpc-03c82286b22e3f973" -> null

# (4 unchanged attributes hidden)

}

# aws_subnet.public_sb will be destroyed

- resource "aws_subnet" "public_sb" {

  - arn = "arn:aws:ec2:ap-south-1:931628309022:subnet/subnet-0b112a29706cfffaf0" ->
null

  - assign_ipv6_address_on_creation = false -> null

  - availability_zone = "ap-south-1c" -> null

  - availability_zone_id = "aps1-az2" -> null

  - cidr_block = "10.0.1.0/24" -> null

  - enable_dns64 = false -> null

  - enable_lni_at_device_index = 0 -> null

  - enable_resource_name_dns_a_record_on_launch = false -> null

  - enable_resource_name_dns_aaaa_record_on_launch = false -> null

  - id = "subnet-0b112a29706cfffaf0" -> null

  - ipv6_native = false -> null

  - map_customer_owned_ip_on_launch = false -> null

  - map_public_ip_on_launch = false -> null

  - owner_id = "931628309022" -> null

  - private_dns_hostname_type_on_launch = "ip-name" -> null

  - region = "ap-south-1" -> null

  - tags = {

    - "Name" = "public_subnet"

  } -> null

```

```

- tags_all                        = {

  - "Name" = "public_subnet"

} -> null

- vpc_id                        = "vpc-03c82286b22e3f973" -> null

# (4 unchanged attributes hidden)

}

# aws_vpc.main will be destroyed

- resource "aws_vpc" "main" {

  - arn                        = "arn:aws:ec2:ap-south-1:931628309022:vpc/vpc-03c82286b22e3f973" -> null

  - assign_generated_ipv6_cidr_block = false -> null

  - cidr_block                  = "10.0.0.0/16" -> null

  - default_network_acl_id      = "acl-0e1c4ba24f5d4b847" -> null

  - default_route_table_id      = "rtb-017b4f84d9239fd26" -> null

  - default_security_group_id   = "sg-0eb4b19336ec47d9f" -> null

  - dhcp_options_id             = "dopt-04b7a6eba4c9859c3" -> null

  - enable_dns_hostnames        = false -> null

  - enable_dns_support          = true -> null

  - enable_network_address_usage_metrics = false -> null

  - id                          = "vpc-03c82286b22e3f973" -> null

  - instance_tenancy            = "default" -> null

  - ipv6_netmask_length         = 0 -> null

  - main_route_table_id        = "rtb-017b4f84d9239fd26" -> null

  - owner_id                    = "931628309022" -> null

  - region                      = "ap-south-1" -> null

  - tags                        = {} -> null

  - tags_all                    = {} -> null

```

(4 unchanged attributes hidden)

}

aws_vpc_security_group_egress_rule.allow_all_traffic_ipv4 will be destroyed

- resource "aws_vpc_security_group_egress_rule" "allow_all_traffic_ipv4" {

- arn = "arn:aws:ec2:ap-south-1:931628309022:security-group-rule/sgr-031000c5697024e92" -> null

- cidr_ipv4 = "0.0.0.0/0" -> null

- id = "sgr-031000c5697024e92" -> null

- ip_protocol = "-1" -> null

- region = "ap-south-1" -> null

- security_group_id = "sg-09e3a590b8372aadd" -> null

- security_group_rule_id = "sgr-031000c5697024e92" -> null

- tags_all = {} -> null

}

aws_vpc_security_group_egress_rule.allow_all_traffic_ipv6 will be destroyed

- resource "aws_vpc_security_group_egress_rule" "allow_all_traffic_ipv6" {

- arn = "arn:aws:ec2:ap-south-1:931628309022:security-group-rule/sgr-0721836483e607067" -> null

- cidr_ipv6 = "::/0" -> null

- id = "sgr-0721836483e607067" -> null

- ip_protocol = "-1" -> null

- region = "ap-south-1" -> null

- security_group_id = "sg-09e3a590b8372aadd" -> null

- security_group_rule_id = "sgr-0721836483e607067" -> null

- tags_all = {} -> null

}

aws_vpc_security_group_ingress_rule.inbound_http will be destroyed

```
- resource "aws_vpc_security_group_ingress_rule" "inbound_http" {  
  - arn          = "arn:aws:ec2:ap-south-1:931628309022:security-group-rule/sgr-074d6c468c6b65cb6" ->  
null  
  - cidr_ipv4    = "0.0.0.0/0" -> null  
  - from_port    = 80 -> null  
  - id          = "sgr-074d6c468c6b65cb6" -> null  
  - ip_protocol  = "tcp" -> null  
  - region      = "ap-south-1" -> null  
  - security_group_id = "sg-09e3a590b8372aadd" -> null  
  - security_group_rule_id = "sgr-074d6c468c6b65cb6" -> null  
  - tags_all     = {} -> null  
  - to_port      = 80 -> null  
}
```

aws_vpc_security_group_ingress_rule.inbound_ssh will be destroyed

```
- resource "aws_vpc_security_group_ingress_rule" "inbound_ssh" {  
  - arn          = "arn:aws:ec2:ap-south-1:931628309022:security-group-rule/sgr-07876b772b53a3de0" -> null  
  - cidr_ipv4    = "0.0.0.0/0" -> null  
  - from_port    = 22 -> null  
  - id          = "sgr-07876b772b53a3de0" -> null  
  - ip_protocol  = "tcp" -> null  
  - region      = "ap-south-1" -> null  
  - security_group_id = "sg-09e3a590b8372aadd" -> null  
  - security_group_rule_id = "sgr-07876b772b53a3de0" -> null  
  - tags_all     = {} -> null  
  - to_port      = 22 -> null  
}
```


Plan: 0 to add, 0 to change, 12 to destroy.

aws_route_table_association.public_RT_association: Destroying... [id=rtbassoc-0bc13bdec86240786]

aws_vpc_security_group_egress_rule.allow_all_traffic_ipv6: Destroying... [id=sgr-0721836483e607067]

aws_subnet.private_sb: Destroying... [id=subnet-0152a2b6e3cef9f06]

aws_vpc_security_group_egress_rule.allow_all_traffic_ipv4: Destroying... [id=sgr-031000c5697024e92]

aws_vpc_security_group_ingress_rule.inbound_ssh: Destroying... [id=sgr-07876b772b53a3de0]

aws_vpc_security_group_ingress_rule.inbound_http: Destroying... [id=sgr-074d6c468c6b65cb6]

aws_instance.my_tf_instance: Destroying... [id=i-0645e7f97a7490fc7]

aws_vpc_security_group_ingress_rule.inbound_http: Destruction complete after 1s

aws_vpc_security_group_egress_rule.allow_all_traffic_ipv4: Destruction complete after 1s

aws_route_table_association.public_RT_association: Destruction complete after 1s

aws_route_table.public_RT: Destroying... [id=rtb-0f3d3cc3ea8ce8cb2]

aws_vpc_security_group_egress_rule.allow_all_traffic_ipv6: Destruction complete after 1s

aws_vpc_security_group_ingress_rule.inbound_ssh: Destruction complete after 1s

aws_subnet.private_sb: Destruction complete after 1s

aws_route_table.public_RT: Destruction complete after 1s

aws_internet_gateway.my_igw: Destroying... [id=igw-01645dcdef08066bf]

aws_instance.my_tf_instance: Still destroying... [id=i-0645e7f97a7490fc7, 00m10s elapsed]

aws_internet_gateway.my_igw: Still destroying... [id=igw-01645dcdef08066bf, 00m10s elapsed]

aws_instance.my_tf_instance: Still destroying... [id=i-0645e7f97a7490fc7, 00m20s elapsed]

aws_internet_gateway.my_igw: Still destroying... [id=igw-01645dcdef08066bf, 00m20s elapsed]

aws_internet_gateway.my_igw: Destruction complete after 27s

aws_instance.my_tf_instance: Still destroying... [id=i-0645e7f97a7490fc7, 00m30s elapsed]

aws_instance.my_tf_instance: Destruction complete after 32s

aws_subnet.public_sb: Destroying... [id=subnet-0b112a29706cffaf0]

aws_security_group.my_sg: Destroying... [id=sg-09e3a590b8372aadd]

aws_subnet.public_sb: Destruction complete after 0s

aws_security_group.my_sg: Destruction complete after 0s

aws_vpc.main: Destroying... [id=vpc-03c82286b22e3f973]

aws_vpc.main: Destruction complete after 1s

Destroy complete! Resources: 12 destroyed.

PS C:\Users\jayan\Terraform\Terraform_practice\terraform_vpc>

2. Create a script to install Apache2

```
[root@ip-10-0-1-119 ec2-user]# cat ec2.sh
```

```
#!/bin/bash
```

```
yum update -y
```

```
yum install -y httpd
```

```
systemctl start httpd
```

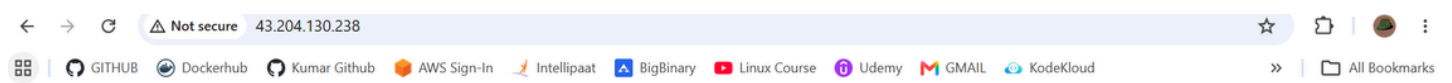
```
systemctl enable httpd
```

```
echo "<html><H1>This content is being served from EC2 instance of VPC created through Terraform</H1>
</html>" > /var/www/html/index.html
```

```
[root@ip-10-0-1-119 ec2-user]#
```

3. Run this script on a newly created EC2 instance

```
./ec2.sh
```



This content is being served from EC2 instance of VPC created through Terraform

4. Print the IP address of the instance in a file on the local once deployed

```
PS C:\Users\jayan\Terraform\Terraform_practice\Terraform_modules> terraform output
```

```
Instance_public_IP = "43.205.123.119"
```

```
PS C:\Users\jayan\Terraform\Terraform_practice\Terraform_modules>
```